Better than Elon Musk - Falcon 9 landing succes

IBM-data-science-specialization
github: https://github.com/snipemuis/IBM-data-science-course/tree/main/capstone_project

Executive Summary

Using open source data the success rate of the Falcon 9 first-stage landing was collected via web scraping and the SpaceX API.

The data was analyzed and presented in a dashboard. The exploratory data analysis identified the best predictors to predict a successful landing. A predictive model was developed to predict if the first-stage Falcon 9 would land. Various models were evaluated with the best performing model having an accuracy of 83%.

Table of Content

1. Introduction
2. methodology
3. Results
4. Conclusions

Introduction

SpaceX Falcon 9 is a rocket that can land the first stage for re-use, significantly reducing costs of a launch.

Not every time the rocket lands and in an unsuccessful landing the first-stage is lost. Thereby increasing the costs of spaceflight.

A new company SpaceY want compete with SpaceX.

The objective of this project is to investigate the success of landing the first stage of the Falcon 9.

To this end the following will be investigates:

- success rate of the Falcon 9 launch / landing
- Identify the best place to make launches
- predicting successful landings of the first stage of rockets

Methodology

Data was obtained from 2 sources

- SpaceX API (https://api.spacexdata.com/v4/rockets/)
- WebScraping (https://en.wikipedia.org/wiki/List_of_Falcon/_9/_and_Falcon_Heavy_launches)

The data was wrangled and a classifier identifying a successful first-stage landing was created

The exploratory data analysis (EDA) was performed using visualization and SQL to identify important features.

Best launch sites were mapped using folium and a dashboard with key information created using Dash.

Predictive analysis was performed by fitting various classification models and evaluated the accuracy of the models.

Methodology: exploratory data analysis

The following SQL queries were performed:

- Names of the unique launch sites in the space mission;
- Top 5 launch sites whose name begin with the string 'CCA';
- Total payload mass carried by boosters launched by NASA (CRS);
- Average payload mass carried by booster version F9 v1.1;
- Date when the first successful landing outcome in ground pad was achieved;
- Names of the boosters which have success in drone ship and have payload mass between 4000 and 6000 kg;
- Total number of successful and failure mission outcomes;
- Names of the booster versions which have carried the maximum payload mass;
- Failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015;
- Rank of the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20.

Methodology: interactive map

Markers, circles, lines and marker clusters were used with Folium Maps

- Markers indicate points like launch sites;
- Circles indicate highlighted areas around specific coordinates, like NASA Johnson Space Center;
- Marker clusters indicates groups of events in each coordinate, like launches in a launch site;
- Lines are used to indicate distances between two coordinates.

Methodology: dashboard

The following graphs and plots were used to visualize data

- Percentage of launches by site
- Payload range
- This combination allowed to quickly analyze the relation between payloads and launch sites

Methodology: classification models

The following models were evaluated

- logistic regression
- support vector machines
- KNN
- decision tree

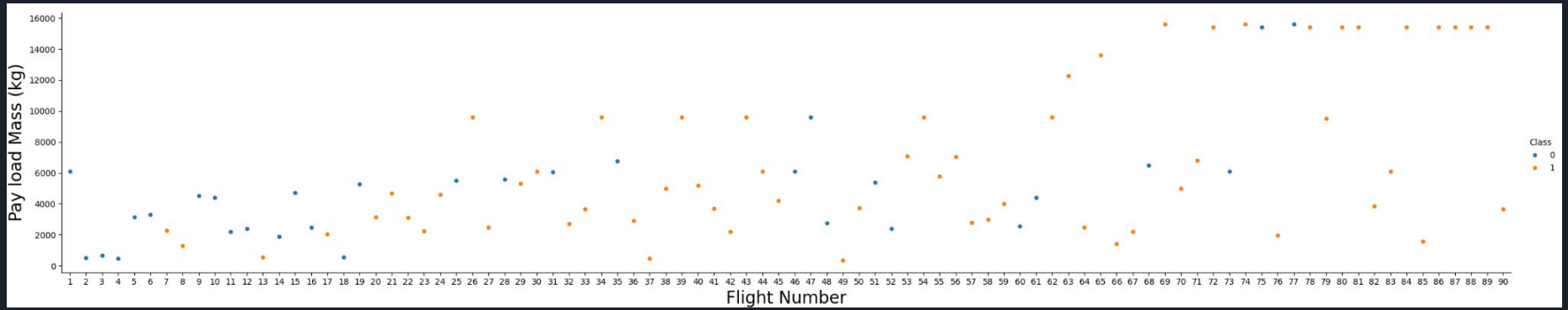
For all models 10-fold cross-validation was performed and gridsearch to identify the best parameters.

Results exploratory data analysis

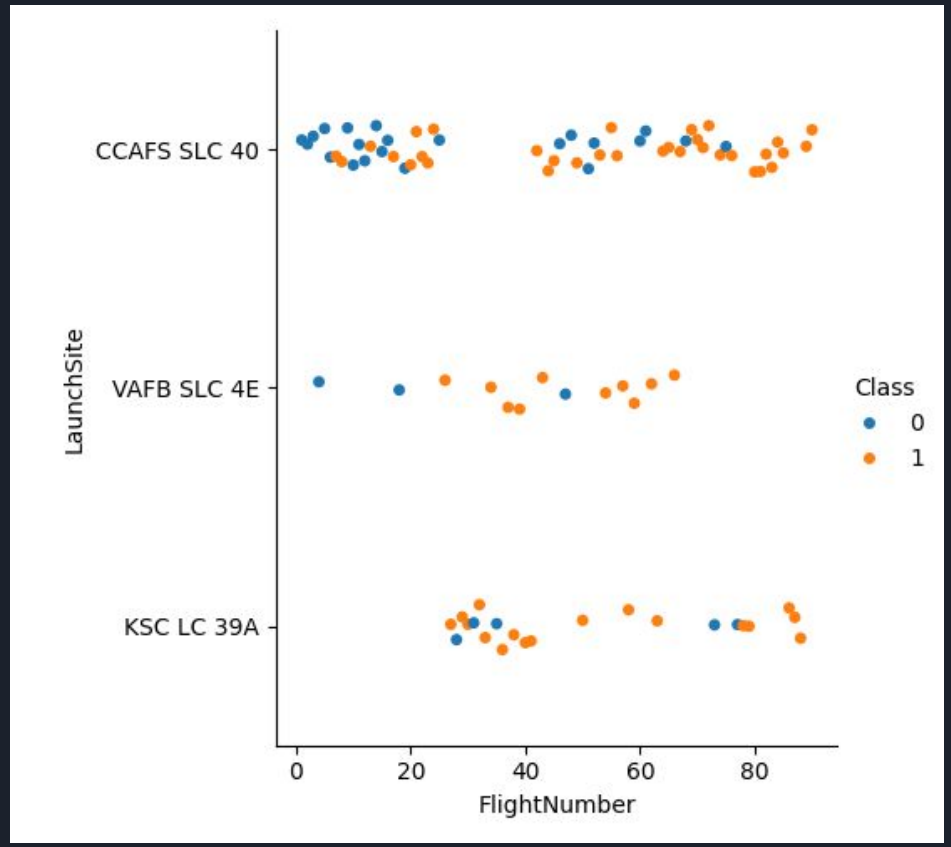
The following SQL queries were performed:

- Space X uses 4 different launch sites;
- The first launches were done to Space X itself and NASA;
- The average payload of F9 v1.1 booster is 2,928 kg;
- The first success landing outcome happened in 2015 fiver year after the first launch;
- Many Falcon 9 booster versions were successful at landing in drone ships having payload above the average;
- Almost 100% of mission outcomes were successful;
- Two booster versions failed at landing in drone ships in 2015: F9 v1.1 B1012 and F9 v1.1 B1015;
- The number of landing outcomes became as better as years passed.

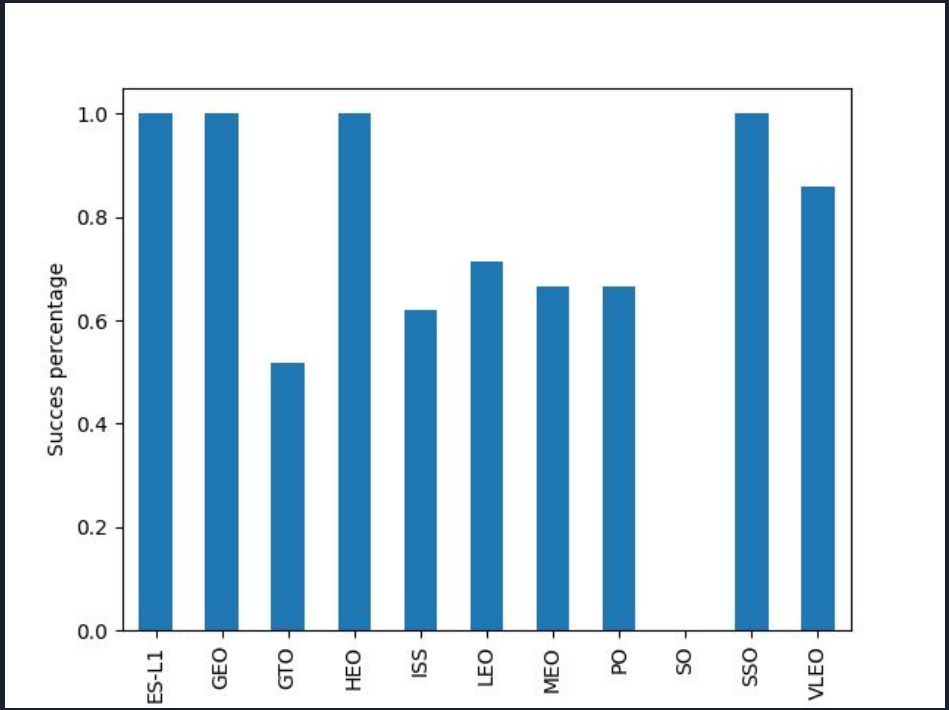
Payload across flightnumbers



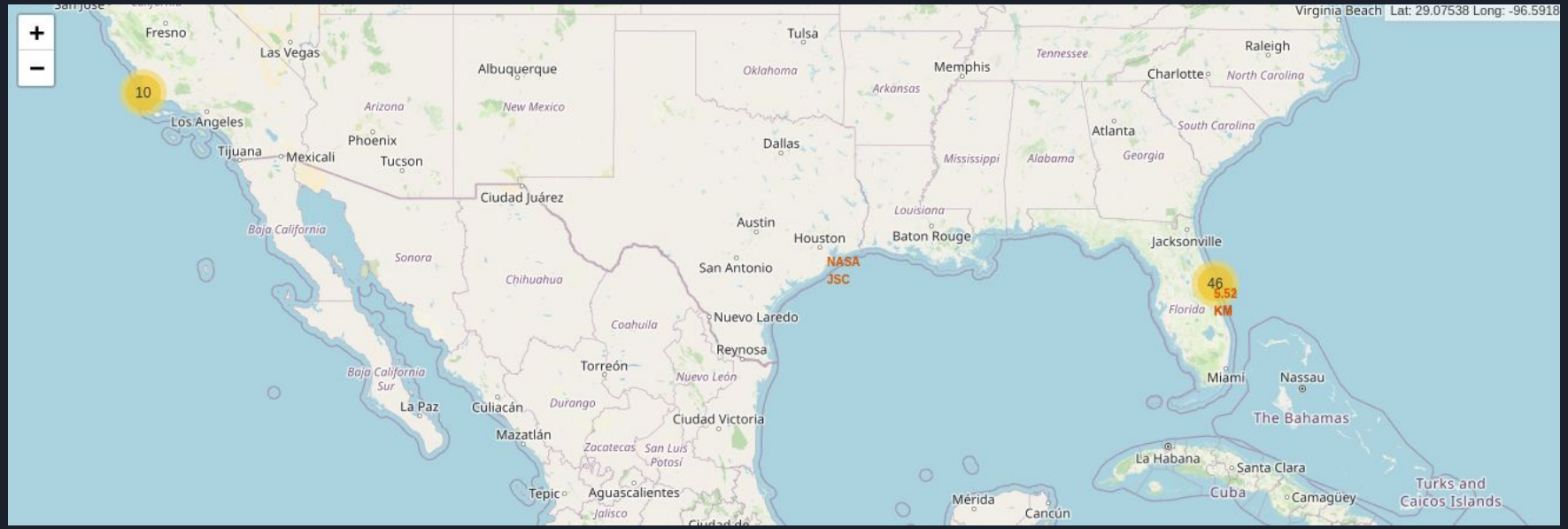
Success percentage across launch sites



Success percentage across Orbits

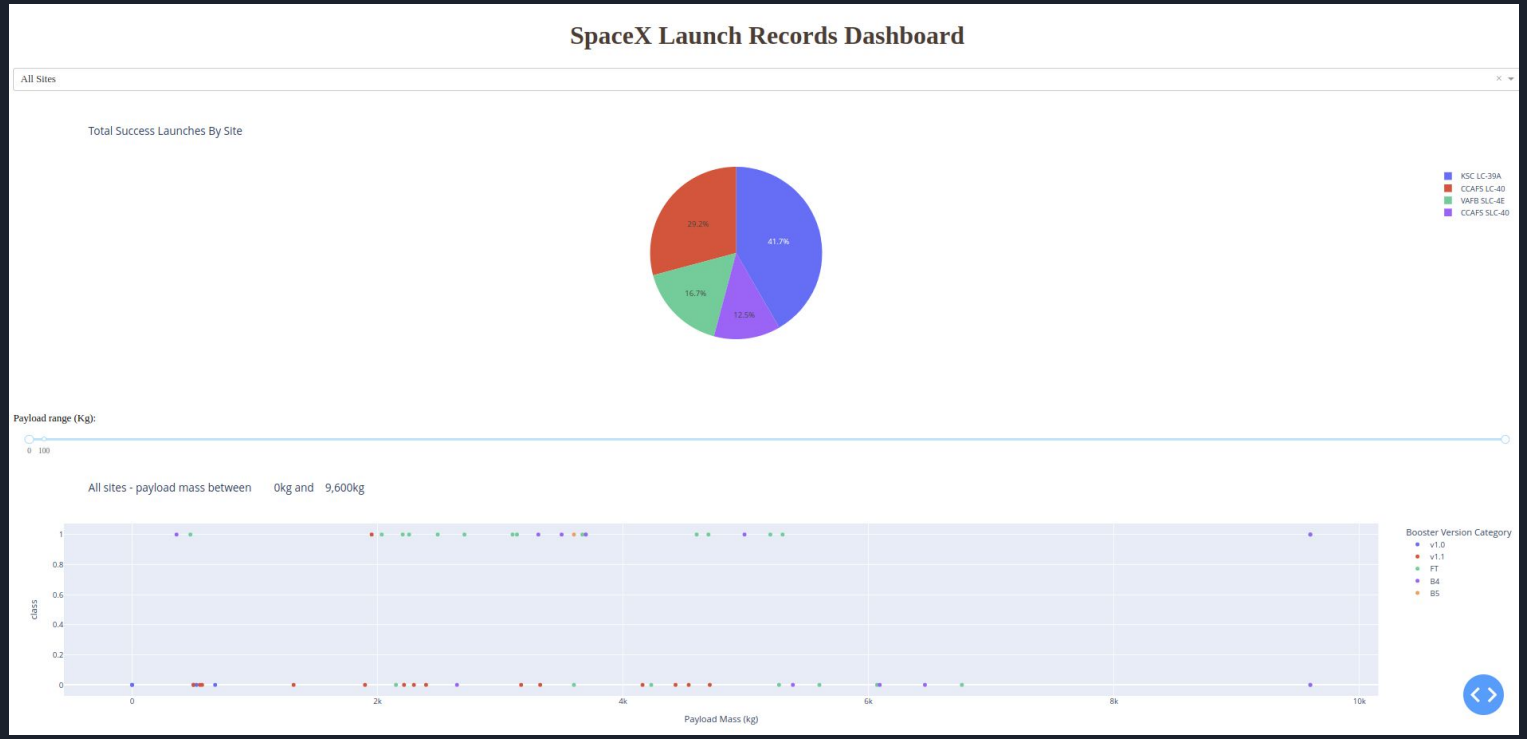


Results interactive map

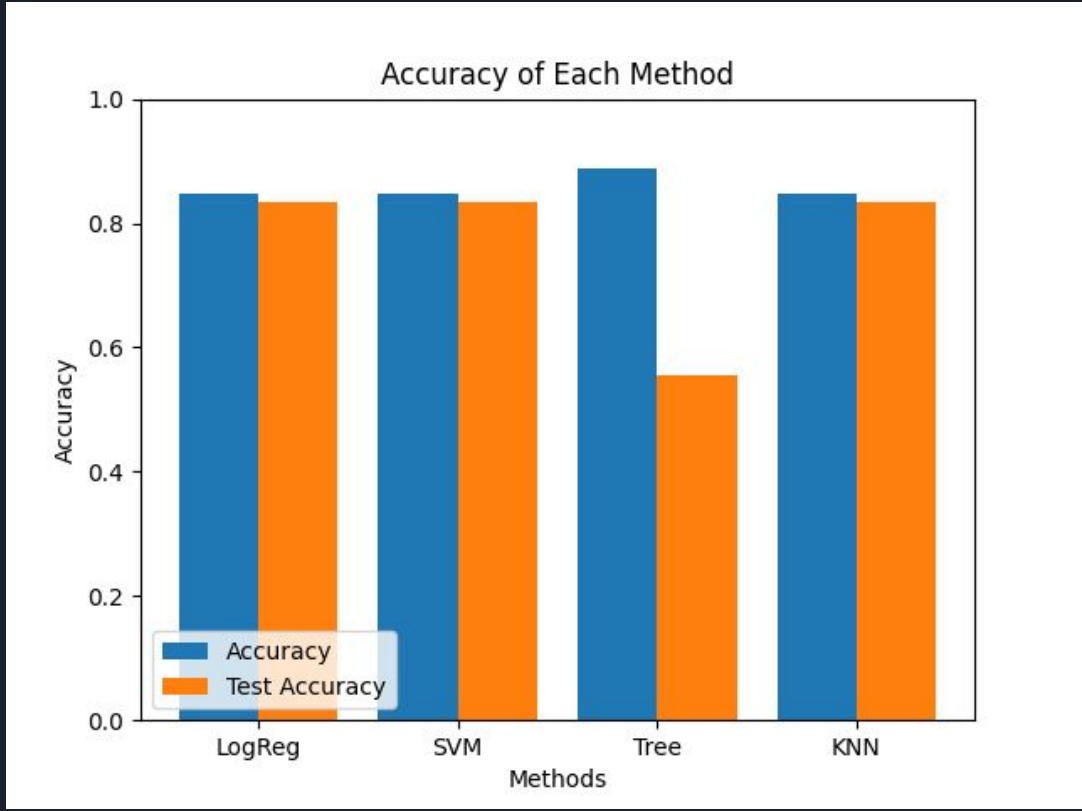


Results interactive map

Results dashboard



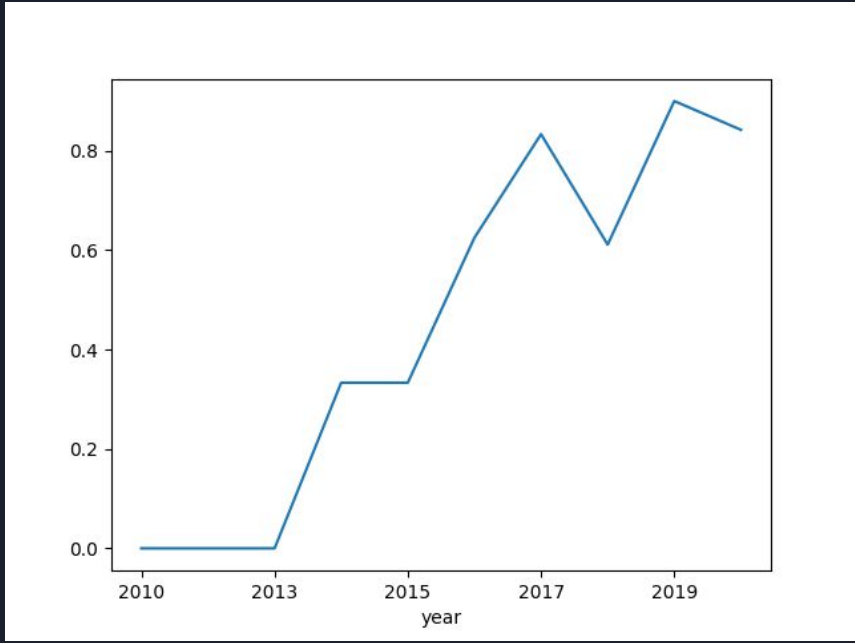
Results classification models



Conclusions

The best launch site is CCAF5 SLC 40 and the success rate for landing the first stage improved over time.

Payloads over 9,000 kg have a high success rate and for payloads over 12,000 kg the best launch sites are CCAFS SLC 40 and KSC LC 39A.

The orbits with the highest success rates are: ES-L1, GEO, HEO, and SSO.

The classification models achieved an accuracy of 83% predicting a successful landing.